

Jiuchen Zhang

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Research interests

Methodology: Statistical and machine learning methods for high-dimensional and longitudinal data; neural networks and data-adaptive models with valid inference

Applications: Exposome modeling and environmental mixtures; life-course and child growth trajectories; environmental health and mobile/wearable health studies

Employment

2024 (July) - **University of Michigan** – Ann Arbor, MI
Present Postdoctoral Research Fellow, Department of Biostatistics
Advisor: Peter X.-K. Song

Education

2020 – 2024 **University of California Irvine** – Irvine, CA
PhD in Statistics
Advisor: Dr. Annie Qu

2016 – 2018 **University of Illinois at Urbana-Champaign** – Champaign, IL
Master of Science in Statistics

2012 – 2016 **Zhejiang University** – Hangzhou, Zhejiang
Bachelor of Science in Mathematics and Applied Mathematics

Methodological & Statistical Research

2026 **Neural Network Machine Regression (NNMR): A Deep Learning Framework for Uncovering High-order Synergistic Effects**
Jiuchen Zhang, Ling Zhou, Peter Song.
arXiv preprint arXiv:2602.02172

- 2026 **Cumulative Treatment Effect Testing under Continuous-Time Reinforcement Learning**
 Jiuchen Zhang, Annie Qu.
arXiv preprint arXiv:2602.02246
- 2024 **Individualized Dynamic Latent Factor Model for Multi-resolutional Data with Application to Mobile Health**
 Jiuchen Zhang, Fei Xue, Qi Xu, Jung-Ah Lee, Annie Qu.
Biometrika, 111(4), 1257–1275
- 2022 **Tensor Factorization Recommender System with Dependency**
 Jiuchen Zhang, Yubai Yuan, Annie Qu.
Electronic Journal of Statistics, 16(1), 2175–2205
- Manuscript in preparation **Function-on-function Regression Model with Multi-resolution and Dependent Predictors**
 Jiming Chen, Jiuchen Zhang, Annie Qu.
- Manuscript in preparation **Likelihood Ratio Test in Binary Tree Classification**
 Jiuchen Zhang, Kai Zhang, Peter Song.

Collaborative & Applied Research

- 2024 **Wearable Internet-of-Things Technology to Measure Sleep of Diverse Caregivers: Preliminary Results from a Randomized Controlled Trial**
 Jung-Ah Lee, Jiuchen Zhang, Amir Rahmani, Annie Qu.
Alzheimer's & Dementia, 20, e092479
- 2023 **Improvements in Heart Rate Variability among Dementia Family Caregivers Receiving a Home-Visit Based Intervention: Preliminary Results from a Randomized Controlled Trial**
 Jung-Ah Lee, Jiuchen Zhang, Eilleen Sabino-Laughlin, Eunae Ju, Amir Rahmani, Annie Qu.
Alzheimer's & Dementia, 19, e077110
- 2022 **Culturally and Linguistically Appropriate Home Visit Intervention for Underserved Dementia Family Caregivers: Preliminary Results from a Randomized Controlled Trial**
 Jung-Ah Lee, Eunae Ju, Jiuchen Zhang, Eilleen Sabino-Laughlin, Amir Rahmani, Lisa Gibbs, Sanghyuk Shin, Annie Qu, Priscilla Kehoe, Adey Nyamathi.
Alzheimer's & Dementia, 18, e058927

In preparation **Identifying Sleep Disturbance Subtypes Among Dementia Caregivers Using Unsupervised Learning**
Eunbee Angela Kim, Jiuchen Zhang, Amir M. Rahmani, Sanghyuk Shin, Adeline Nyamathi, Jung-Ah Lee.

Award

- 2024 First Place, ASA MDD Student Paper Competition (JSM 2024)
- 2024 ENAR Distinguished Student Paper Award (ENAR Spring Meeting)
- 2013 Second-Class Scholarship for Elite Student in Basic Sciences (Zhejiang University)
- 2013 Third-Class Scholarship for Ourstanding Merits (Zhejiang University)

Academic appointment

- Teaching Assistant**, University of California, Irvine
- 2021 Fall STATS 200AP: Intermediate Probability and Statistical Theory I
 - 2021 Spring STATS 7: Basic Statistics

Mentoring

- 2023 Haoqun Cao, Renmin University
Xieqing Yu, Cornell University
- 2022 Zhuojiao Dong, University of California, Irvine
Yiting Wang, University of California, Irvine
Yiheng Wei, University of California, Irvine
Haochen Tan, University of California, Irvine
Ruxi Zheng, University of California, Irvine

Prior Professional Experience

- Dec. 2021 – Jun. 2024 **Research Assistant** – Irvine, CA
Contributed to the Dementia Family Caregiver Support Study through end-to-end management and analysis of wearable device data from smartwatches and smart rings.
Developed automated pipelines for large-scale data downloading and preprocessing in collaboration with engineering teams and computer science researchers.
Conducted statistical analyses supporting multiple conference abstracts and peer-reviewed manuscripts.

- May 2019 – **Statistical Consultant, Illinois Statistics Office** – Champaign, IL
 Dec. 2019 Provided statistical consultation for interdisciplinary grant proposals, including power analysis, study design, and selection of appropriate statistical methodologies. Prepared data management plans and statistical reports; conducted quality control and code review for biomedical data analyses using R, Python, and SAS.
- Sep. 2018 – **Statistical Analyst, Technical Consulting & Research, Inc.** – Weston, CT
 May 2019 Developed Python-based data acquisition pipelines using Selenium and BeautifulSoup to extract and integrate clinical trial information across multiple sources. Performed data cleaning, record linkage, and geocoding to harmonize trial records with multiple locations and contacts.

Talks & Conference Service

- Aug. 2025 A Scaling Neural Network-Based Approach for Variable Selection in Nonparametric Regression
2025 Joint Statistical Meetings (JSM)
- Aug. 2025 Harnessing the Power of Large-Scale and Heterogeneous Data with Integrative Analysis
Session Chair, *2025 Joint Statistical Meetings (JSM)*
- Aug. 2024 Individualized Dynamic Latent Factor Model for Multi-resolutional Data with Application to Mobile Health
2024 Joint Statistical Meetings (JSM)
- Mar. 2024 Individualized Dynamic Latent Factor Model for Multi-resolutional Data with Application to Mobile Health
2024 Eastern North American Region (ENAR) Spring Meeting
- Aug. 2023 Individualized Dynamic Model for Multiresolutional Data
2023 Joint Statistical Meetings (JSM)
- Sep. 2021 A Tensor Factorization Recommender System with Dependency
Invited talk, 2021 ICSA Applied Statistics Symposium
- Aug. 2020 Multilayer Recommender Systems Using Dependent Tensor
2020 Joint Statistical Meetings (JSM)

Referee Experience

Electronic Journal of Statistics (3 reviews)

Journal of the American Statistical Association (3 reviews)

Journal of the Royal Statistical Society: Series B (3 reviews)

ACM Transactions on Knowledge Discovery from Data (1 review)

Scientific Reports (1 review)

Professional Membership

American Statistical Association

International Chinese Statistical Association

Institute of Mathematical Statistics

Eastern North American Region International Biometric Society